



BERLIN SCHOOL OF ECONOMICS MASTER CLASS 2026

Foundations of Difference-in-Differences

Day 1: 2×2 , Parallel Trends, and Covariates

Scott Cunningham · Baylor University

Berlin School of Economics · Master Class 2026

About me

- **Scott Cunningham** — Professor of Economics, Baylor University; Visiting Professor of Government, Harvard University.
- This is a **3-day workshop on difference-in-differences** — from the basic 2×2 through continuous treatment and staggered adoption.
- Much of the material is drawn from **Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (2026)**, “Difference-in-Differences: A Practitioner’s Guide,” *Journal of Economic Literature*, and new work on continuous-treatment DiD.

Three days

Day	Time	Topic
Day 1	10:00–11:30am	2×2 DiD: parameters, calculations, identification, covariates
Day 2	9:00–10:30am	Continuous treatment DiD
Day 3	11:00am–12:30pm	Staggered adoption DiD

Today: from the 2×2 to covariates

- What the 2×2 DiD is — as a **calculation** and as an **identification strategy**.
- Six equivalent ways to compute the 2×2 — and why that matters.
- What the DiD actually identifies: ATT under parallel trends.
- Event studies — what they tell you and what they don't.
- When covariates are needed and how to include them correctly.

LESSON

DiD is Two Things: Calculation + Assumption

DiD is a research design, not a regression

A research design is an identification strategy: it reveals the invisible target parameter hidden inside a calculation.



The individual treatment effect

$$\delta_i = Y_i(1) - Y_i(0)$$

- One per unit. **Different** across units in general — treatment effects are heterogeneous.
- We only ever see one of $Y_i(1)$ or $Y_i(0)$. The other is the counterfactual.
- So δ_i is never directly observed at the individual level.

The ATT: a causal parameter has three ingredients

$$\text{ATT} = \underbrace{\mathbb{E}}_{\text{weighted expectations}} \left[\underbrace{Y(1) - Y(0)}_{\text{potential outcomes}} \mid \underbrace{D = 1, \text{Post}}_{\text{population}} \right]$$

- **Potential outcomes** — the contrast $Y(1) - Y(0)$
- **Population** — on whom: treated units, post-period
- **Weighted expectations** — how we average across that population

The switching equation: you see one half

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

- For each unit: $Y_i(1)$ if treated, $Y_i(0)$ if not.
- The ATT asks about *both* columns. The data hands you one.

The rest of today: strategies for recovering the ATT from the half we see.

LESSON

Six Equivalent Ways, One Number

Cheng & Hoekstra (2013): castle doctrine and homicide

The reform. “Castle doctrine” laws expanded the legal right to use lethal force in self-defense. Many US states passed expansions from **2005–2010**.

- **Panel:** all US states $\times$ years. **Outcome:** log homicide rate.
- **Treatment:** state passed a castle doctrine expansion.
- **Comparison:** states that did not pass (never-treated).

For today: a clean 2×2 on **2005–2006**. Treated = states adopting in 2006; control = states that hadn't. *Cheng & Hoekstra (2013), J. Human Resources 48(3): 821–854.*

The four cells, by hand on castle.dta

```
use castle.dta, clear
keep if year >= 2005 & year <= 2006
gen post = (year == 2006)
gen treat = (effyear == 2006)

summarize l_homicide if treat==1 & post==1 // Y11
summarize l_homicide if treat==1 & post==0 // Y10
summarize l_homicide if treat==0 & post==1 // Y01
summarize l_homicide if treat==0 & post==0 // Y00
```

$$\widehat{\text{DiD}} = (\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00}) \approx \mathbf{0.10}$$

Log homicide rate ~ 10% higher in 2006-adopting states relative to non-adopters.

Four averages, three subtractions — two orderings

The 2×2 is four cell means: $\bar{Y}_{11}$, $\bar{Y}_{10}$, $\bar{Y}_{01}$, $\bar{Y}_{00}$.

First-difference ordering ($\rightarrow$)

Subtract across time *within* each group, then compare groups:

$$\Delta_1 = \bar{Y}_{11} - \bar{Y}_{10}$$

$$\Delta_0 = \bar{Y}_{01} - \bar{Y}_{00}$$

$$\text{DiD} = \Delta_1 - \Delta_0$$

Group-difference ordering ($\downarrow$)

Subtract across groups *within* each period, then compare periods:

$$G_{\text{post}} = \bar{Y}_{11} - \bar{Y}_{01}$$

$$G_{\text{pre}} = \bar{Y}_{10} - \bar{Y}_{00}$$

$$\text{DiD} = G_{\text{post}} - G_{\text{pre}}$$

Same four averages. Same three subtractions. Same number. The 2×2 is symmetric under reordering.

Six equivalent ways, one number

Way	What it does	$\hat{\beta}_{DD}$
1. First-difference arithmetic	$(\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})$	≈ 0.10
2. Group-difference arithmetic	$(\bar{Y}_{11} - \bar{Y}_{01}) - (\bar{Y}_{10} - \bar{Y}_{00})$	≈ 0.10
3. OLS with <code>post##treat</code>	saturated dummies, direct interaction	≈ 0.10
4. Two-way fixed effects (TWFE)	absorbs unit + time means	≈ 0.10
5. ΔY_i on D_i (horizontal)	time-difference within group, then on D	≈ 0.10
6. G_t on Post_t (vertical)	group-difference within period, then on Post	≈ 0.10

Same data, six ways, identical to machine epsilon.

“DiD” is not a regression — there are six equivalent ways to compute it

- You will often hear: “I ran a diff-in-diff regression.”
- That is misleading. Two of the six ways are pure arithmetic; four are regressions. All give the same answer.
- The regressions are tools for requesting the estimate. The **2×2 is the thing being estimated**.

This is why we start with the 2×2, not the regression. The estimand is four averages and three subtractions. The regression is just a convenient way to compute — and standard-error — it.

LESSON

Decomposing the 2×2

The proof: add a zero, rearrange (four steps)

1. Start with the 2×2 :

$$\widehat{\text{DiD}} = [\mathbb{E}[Y \mid 1, \text{Post}] - \mathbb{E}[Y \mid 1, \text{Pre}]] - [\mathbb{E}[Y \mid 0, \text{Post}] - \mathbb{E}[Y \mid 0, \text{Pre}]]$$

2. Switching equation. Treated-post is $Y(1)$; the other three cells are $Y(0)$:

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}]] - [\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]]$$

3. Add a zero. Insert $+\mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}]$ (the missing counterfactual, twice):

$$\begin{aligned} \widehat{\text{DiD}} &= \mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Post}] \\ &\quad + \mathbb{E}[Y(0) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}] - [\mathbb{E}[Y(0) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]] \end{aligned}$$

4. Rearrange. First two terms = ATT; remainder = difference in untreated trends:

$$\widehat{\text{DiD}} = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid 1, \text{Post}]}_{\text{ATT}} + \underbrace{\Delta \mathbb{E}[Y(0) \mid 1] - \Delta \mathbb{E}[Y(0) \mid 0]}_{\text{PT bias} = 0 \text{ under parallel trends}}$$

The bias term *is* selection bias — on trends, not levels

$$\text{bias} = \underbrace{\Delta \mathbb{E}[Y(0) \mid D=1]}_{\text{how treated's } Y(0) \text{ moves}} - \underbrace{\Delta \mathbb{E}[Y(0) \mid D=0]}_{\text{how controls' } Y(0) \text{ moves}}$$

- **Cross-section selection bias:** are treated and control different in *levels* of $Y(0)$?
- **DiD's selection bias:** are treated and control different in *trends* of $Y(0)$?
- Different levels are fine — unit fixed effects absorb them.
- What can't differ is the *rate* at which $Y(0)$ moves across groups.

The trade DiD makes vs. randomization

RCTs guarantee identical trends (and levels) through the design itself.

DiD requires only one thing: parallel trends in $Y(0)$. But unlike randomization, **there is no guaranteed research design that delivers it** — it must be argued for on substantive grounds.

Ghanem, Sant'Anna & Wüthrich (2025).

Case 2: no-anticipation fails — treated group adjusts early

Treated units *anticipate* the policy and adjust before it lands. Their pre-period outcome is contaminated:

$$\mathbb{E}[Y \mid 1, \text{Pre}] = \mathbb{E}[Y(0) \mid 1, \text{Pre}] + \text{ATT}_{\text{pre}}$$

Apply the same +0 trick as Case 1. The contaminated pre-period subtracts off:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}_{\text{Post}}}_{\text{what we want}} + \underbrace{\Delta \mathbb{E}[Y(0) \mid 1] - \Delta \mathbb{E}[Y(0) \mid 0]}_{\text{PT bias} = 0} - \underbrace{\text{ATT}_{\text{pre}}}_{\text{anticipation bias}}$$

Numerical example

$$\text{ATT} = 10, \text{PT bias} = 0, \text{ATT}_{\text{pre}} = 3. \quad \widehat{\text{DiD}} = 10 - 3 = 7. \quad \text{Biased down.}$$

If effects are constant over time, $\text{ATT}_{\text{pre}} = \text{ATT}_{\text{post}}$ and $\text{DiD} \rightarrow 0$.

Case 3: SUTVA fails — control group is also treated in Post

The comparison group's Post outcome is $Y(1)$, not $Y(0)$ (spillover or simultaneous treatment):

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1) \mid 1, \text{Post}] - \mathbb{E}[Y(0) \mid 1, \text{Pre}]] - [\mathbb{E}[Y(1) \mid 0, \text{Post}] - \mathbb{E}[Y(0) \mid 0, \text{Pre}]]$$

Add *two* zeros (one counterfactual on each side) and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}}_{\text{treated}} + \underbrace{\Delta \mathbb{E}[Y(0) \mid 1] - \Delta \mathbb{E}[Y(0) \mid 0]}_{\text{PT bias} = 0} - \underbrace{\text{ATT}_{D=0, \text{Post}}}_{\text{control's treatment effect}}$$

Numerical example

$\text{ATT} = 10$, $\text{PT bias} = 0$, $\text{ATT}_{D=0, \text{Post}} = 4$. $\widehat{\text{DiD}} = 10 - 4 = 6$. **Biased toward zero.**

Case 4: comparison group is always treated

The comparison group was treated *before* our data starts. Both their Pre and Post outcomes are $Y(1)$:

$$\widehat{\text{DiD}} = [\mathbb{E}[Y(1) | 1, \text{Post}] - \mathbb{E}[Y(0) | 1, \text{Pre}]] - [\mathbb{E}[Y(1) | 0, \text{Post}] - \mathbb{E}[Y(1) | 0, \text{Pre}]]$$

Add $\pm \mathbb{E}[Y(0) | 0, \text{Post}]$ and $\pm \mathbb{E}[Y(0) | 0, \text{Pre}]$ on the comparison side and rearrange:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}}_{\text{treated}} + \underbrace{\Delta \mathbb{E}[Y(0) | 1] - \Delta \mathbb{E}[Y(0) | 0]}_{\text{PT bias} = 0} - \underbrace{\Delta \text{ATT}_{D=0}}_{\text{change in comparison's TE}}$$

Numerical example

$\text{ATT} = 10$, $\text{PT bias} = 0$, $\Delta \text{ATT}_{D=0} = 2$. $\widehat{\text{DiD}} = 10 - 2 = 8$. **Biased toward zero.**

Four cases: one method

Case	What fails	$\widehat{\text{DiD}} =$
1. NA + untreated comparison	nothing (baseline)	ATT + PT bias
2. NA fails	treated Pre $\supset Y(1)$	ATT + PT bias – ATT_{pre}
3. Spillover (SUTVA)	control Post $\supset Y(1)$	ATT + PT bias – $\text{ATT}_{D=0, \text{Post}}$
4. Always-treated comparison	control Pre & Post $\supset Y(1)$	ATT + PT bias – $\Delta \text{ATT}_{D=0}$

Same method every time: write the 2×2 , apply the switching equation, add the right zero(s), rearrange. Only what you add changes.

New Jersey raises its minimum wage. Pennsylvania doesn't.

- **April 1992**: NJ raises its minimum wage from \$4.25 to \$5.05 — a 19% increase.
- Pennsylvania, just over the river, stays at \$4.25.
- Card and Krueger survey 410 fast-food restaurants on both sides of the border, before (Feb 1992) and after (Nov 1992).
- The natural 2×2 : **treated = NJ, control = PA, pre/post the policy.**

Card & Krueger, American Economic Review, 1994.

The 2×2 as long differences: horizontal regression

Step 1: Within each group, subtract across time (pre → post).

	Pre	Post	$\Delta (\rightarrow)$
NJ (treated)	20.44	21.03	+0.59
PA (control)	23.33	21.17	-2.16

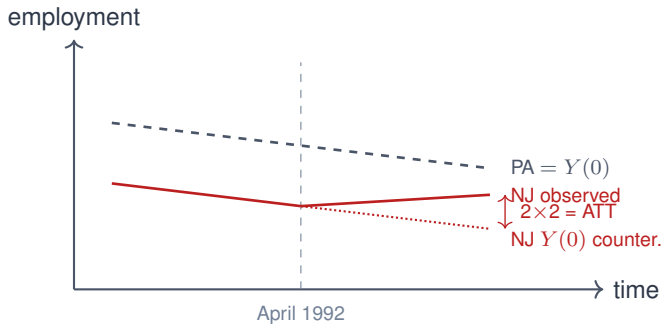
Step 2: Subtract the two time changes:

$$\hat{\beta}_{\text{DiD}} = (+0.59) - (-2.16) = \mathbf{+2.75}$$

Imbens: the **long-difference regression** (or **horizontal regression**)

$\Delta Y_i = \alpha + \beta_{\text{DiD}} D_i + u_i$ — difference over time *within* each group, then compare groups.

Parallel trends visualized: the control *is* the counterfactual



The dotted line is what NJ would have done absent the policy. We never see it.

PT lets PA's slope stand in for it.

The same 2×2 as group differences: vertical regression

Step 1: At each time period, subtract across groups (NJ–PA, ↓).

	Pre	Post
NJ (treated)	20.44	21.03
PA (control)	23.33	21.17
Gap (NJ–PA ↓)	–2.89	–0.14

Step 2: Subtract the two group gaps:

$$\hat{\beta}_{\text{DiD}} = (-0.14) - (-2.89) = \mathbf{+2.75}$$

Imbens: the **group-differences regression** (or **vertical regression**)

$D_t = \alpha + \beta_{\text{DiD}} \text{Post}_t + v_t$ — compare groups *within* each period, then difference across time.

Two orderings, one number — and why the vertical view matters

Ordering	Regression	Estimate
Time first, groups second ($\rightarrow$ then $\uparrow$)	Long-difference (horizontal)	+2.75
Groups first, time second ($\downarrow$ then $\rightarrow$)	Group-differences (vertical)	+2.75

- Both orderings of the double-difference give the same number.
- The 2×2 is symmetric under reordering — it doesn't matter which subtraction comes first.

Why the vertical view matters (Imbens 2021): Allow an intercept in the group-differences regression: $D_t = \mu + \beta_{\text{DiD}} \text{Post}_t$. That μ *absorbs the level difference between the two groups* — which is the key move in synthetic control. No convex-hull constraint required.

Three flavors of standard error

- **Vanilla (homoskedastic)**: assumes $\text{Var}(\varepsilon_i | X_i) = \sigma^2$ for all i .
- **Robust (HC)**: each residual speaks for itself — handles heteroskedasticity.
- **Cluster-robust (CRVE)**: groups residuals by cluster; within-cluster errors correlate freely.

Same point estimate, three different SEs. The SE you pick is a claim about the error structure.

Bertrand, Duflo & Mullainathan (QJE 2004): the placebo-law test

“How Much Should We Trust Differences-in-Differences Estimates?”

- BDM generate *placebo laws*: assign random treatment dates to states in a real CPS panel.
- True effect by construction: $\beta = 0$. So $\Pr(\text{reject } H_0)$ should equal $\alpha = 5\%$.
- **Vanilla SE**: rejection rate $\approx 45\%$. Wildly over-rejects.
- **HC-robust**: rejection rate *also* $\approx 45\%$. **HC barely moves the needle.**
- **Cluster-robust at state level**: rejection rate $\approx 5\%$. The fix.

Why HC doesn't save you

Binary, balanced D + within-state serial correlation. The het-correction has nothing to do with the actual problem. **Clustering is doing all the work.**

LESSON

Five Pieces of a Strong DiD

Five pieces of a strong DiD

1. **Bite** — the policy actually moved what it was supposed to move.
2. **Event study** — pre/post dynamics around the treatment date.
3. **Falsification** — a rival theory fails on a placebo group or outcome.
4. **Main result** — the DiD on the outcome that matters.
5. **Mechanism** — the causal pathway from D to Y .

We will walk through all five using a single case study: Medicaid expansion and near-elderly mortality.

From 2×2 to $2 \times T$: the event study regression

Extend from one pre/post comparison to a full timeline:

$$Y_{it} = \alpha_i + \tau_t + \sum_{k \neq -1} \beta_k \cdot (D_i \cdot \mathbf{1}\{t = k\}) + \varepsilon_{it}$$

- α_i, τ_t — unit and time fixed effects. $t = -1$ is omitted (the reference period).
- Each $\hat{\beta}_k$ is a 2×2 DiD against $t = -1$: $\hat{\beta}_k = (\bar{Y}_k^T - \bar{Y}_{-1}^T) - (\bar{Y}_k^C - \bar{Y}_{-1}^C)$.
- Pre-period ($k < -1$): placebo estimates — should be near zero under PT.
- Post-period ($k \geq 0$): dynamic treatment effects.

The event study gives evidence, not identification

The 2×2 estimator recovers:

$$\widehat{\text{DiD}} = \underbrace{\text{ATT}}_{\text{target}} + \underbrace{\left(\mathbb{E}[\Delta Y(0) \mid D=1] - \mathbb{E}[\Delta Y(0) \mid D=0] \right)}_{\text{PT bias}}$$

- **PT bias** = the gap between treated and control *untreated* trends.
- Parallel trends *assumes* this is zero — we never observe $Y(0)$ for the treated post-treatment.
- Pre-period placebo coefficients ($\hat{\beta}_k, k < -1$) ask: was this bias near zero *before*?

An event study is **not** needed for identification — PT is an assumption, not a testable claim.

It is needed for **evidence**: flat pre-trends suggest the bias was small in the past, which makes PT plausible going forward.

Medicaid expansion and near-elderly mortality

Miller, Johnson & Wherry (2021), *Quarterly Journal of Economics*.

- Did the ACA Medicaid expansion (2014) reduce mortality among the **near-elderly** (ages 55–64)?
- Comparison: states that expanded Medicaid vs. states that didn't, before vs. after 2014.
- Headline finding: $\approx 9.2\%$ **reduction** in near-elderly mortality.

Why *near-elderly*?

Everyone 65+ already has Medicare — Medicaid expansion is irrelevant for them. The elderly become a natural **placebo group** living in the same states and the same macroeconomy.

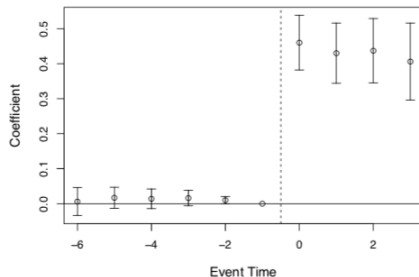
Strategy: walk through all five pieces of a strong DiD, in order.

Five pieces — start with bite

1. **Bite** — the policy actually moved what it was supposed to move.
2. *Event study*
3. *Falsification*
4. *Main result*
5. *Mechanism*

If the policy didn't move the treatment variable, nothing downstream is credible.

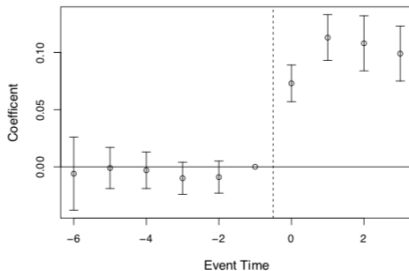
Bite #1: Medicaid take-up jumped in expansion states



(a) Medicaid Eligibility

Did the policy move people onto Medicaid? Yes — enrollment rose sharply in expansion states post-2014, flat in non-expansion states. If the law didn't actually do something, nothing downstream is attributable to it.

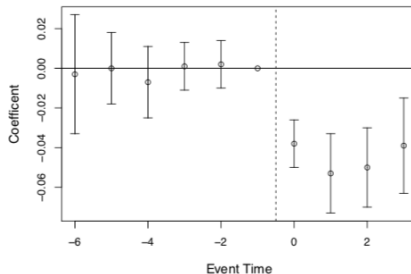
Bite #2: any-insurance coverage also rose



(b) Medicaid Coverage

Did Medicaid crowd out private insurance, or did total coverage rise? Coverage rose broadly in expansion states — take-up is real, not mere substitution from one insurer to another.

Bite #3: the uninsured rate fell in expansion states



(c) Uninsured

Third measure, same story. *Uninsured rate dropped sharply in expansion states post-2014 and stayed flat in non-expansion states. Three independent measures of bite all point the same direction.*

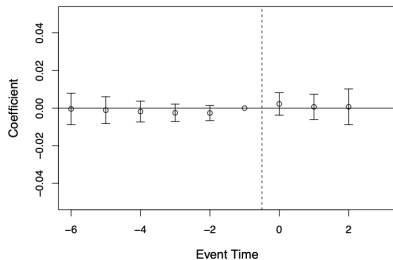
Five pieces — now falsify

1. *Bite* ✓
2. *Event study*
3. **Falsification** — a rival theory fails on a placebo group or outcome.
4. *Main result*
5. *Mechanism*

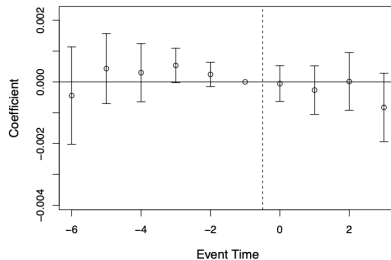
A rival explanation: some 2014 shock hit expansion states. If true, it would also hit the 65+ population. It didn't.

Falsification: same model, 65+ population — no effect

Age 65+ in 2014



(c) Medicaid Coverage



(d) Annual Mortality

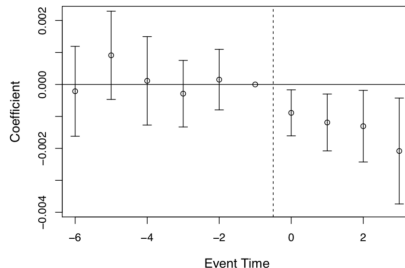
Same states, same years, same model — but run on the 65+ population. They all have Medicare, so Medicaid expansion shouldn't touch their mortality. A rival story ("some 2014 shock hit expansion states") would predict a mortality effect here. It didn't appear. **Rival rejected.**

Five pieces — event study and main result

1. *Bite* ✓
2. **Event study** — pre/post dynamics around the treatment date.
3. *Falsification* ✓
4. **Main result** — the DiD on the outcome that matters.
5. *Mechanism*

Flat pre-trends build credibility. The post-2014 break is the result.

Main result: $\approx 9.2\%$ reduction in near-elderly mortality



- Post-2014: mortality declines in expansion states. $\approx 9.2\%$ **reduction**.
- Pre-period $\hat{\beta}_k$ near zero. Wide CIs — but the event study + falsification + bite tell the full story.
- A critic must now explain bite *and* falsification *and* the dynamic post-trend simultaneously.

Five pieces — mechanism

1. *Bite* ✓
2. *Event study* ✓
3. *Falsification* ✓
4. *Main result* ✓
5. **Mechanism** — the causal pathway from D to Y .

What kind of mortality falls? Disease-specific decomposition tells us where Medicaid is operating.

Mechanism: disease-related deaths fall

- Mechanisms are the part of a result that is **transportable** to other settings and populations.
- Reduced-form results often bundle several mechanisms — decompositions separate them.
- Here: **disease-related deaths fall** (cardiovascular, respiratory, infectious disease); injury deaths do not.
- Consistent with **chronic-disease management** — exactly what insurance-mediated access to care would facilitate.

Mechanisms tell you *why* it worked and *how far the result generalizes*. If the mechanism is access to preventive care, the finding transfers to other populations with similar unmet preventive needs.

The five pieces, all five present in Medicaid

Piece	Test	Result
Bite	Enrollment, any-insurance, uninsured rate	All three move
Event study	Pre- $\hat{\beta}_k$ near zero; 2014 break	Flat pre-trends
Falsification	Same model, 65+ population (Medicare)	No effect
Main result	DiD on near-elderly mortality	-9.2%
Mechanism	Disease-specific decomposition	Chronic disease

A critic must explain all five simultaneously. The evidentiary burden is now very high.

LESSON

Covariates in DiD

A thought experiment

Treatment = college education. Outcome = wages. Confounder = sex.

Group	Untreated wage trend $\Delta Y(0)$
Male workers	+10 per year
Female workers	+8 per year

Case 1: $D=1$ is 75% male. $D=0$ is 75% male.

PT bias = $9.5 - 9.5 = 0$. ✓ Parallel trends holds.

Case 2: $D=1$ is 75% male. $D=0$ is **25%** male.

PT bias = $9.5 - 8.5 = 1.0 \neq 0$. $\hat{\delta}_{\text{DiD}} \neq \text{ATT}$.

Why? Imbalance on a trend-causing covariate

DiD does **not** require the two groups to look the same on *levels*.

DiD **does** require the same $Y(0)$ *trends*.

Imbalance in a covariate that *causes* $Y(0)$ trends will **mechanically** break parallel trends.

Fix: condition on sex. Several ways:

1. TWFE saturated regressions
2. Inverse probability weighting (IPW)
3. Outcome regression (OR)
4. Doubly robust (DR)

For conditioning to work: conditional PT

We need parallel trends *within* each group:

1. Male treated workers have the *same* $Y(0)$ trend as male control workers
2. Female treated workers have the *same* $Y(0)$ trend as female control workers

Conditional PTA: $\mathbb{E}[\Delta Y(0) \mid D=1, X] = \mathbb{E}[\Delta Y(0) \mid D=0, X]$ for all X

Parallel trends is now an assumption you're making *within cells defined by X* .

The standard TWFE with additive X — Bias #1

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \varepsilon_{it}$$

Hidden assumption: returns to X are constant over time ($\beta_{\text{pre}} = \beta_{\text{post}}$).

PT bias when this fails:

$$\underbrace{E[\Delta Y(0) \mid D=1] - E[\Delta Y(0) \mid D=0]}_{\text{PT bias}} = \Delta\beta \cdot (E[X \mid D=1] - E[X \mid D=0])$$

PT fails whenever returns shift ($\Delta\beta \neq 0$) **and** groups are imbalanced on X .

Fix for Bias #1: interact Post $\times$ X

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \gamma (\text{Post}_t \times X_i) + \varepsilon_{it}$$

γ absorbs any shift in returns to X between pre and post.

```
reg re i.post##i.ever_treated ///  
    i.post##c.age i.post##c.agesq i.post##c.agecube ///  
    i.post##c.educ i.post##c.educsq ///  
    i.post##i.marr i.post##i.nodegree i.post##i.black i.post##i.hisp ///  
    i.post##c.re74 i.post##i.u74, robust
```

Three biases — one family of fixes

Bias	What TWFE imposes	Fix
#1 Trend imbalance	β constant pre and post	$\text{Post} \times X$
#2 Heterogeneous TE	δ same at every X	$D \times X$ (saturate treatment)
#3 Time-invariant X	X_i survives demeaning	$\text{Post} \times X$ (same fix as #1)

Most researchers include additive X and call it a day. That is the wrong spec. To fix all three you need both $\text{Post} \times X$ and $D \times X$. It is rare to see this in published work.

LESSON

Alternative Estimators: IPW, OR, DR

IPW: reweight the controls to look like the treated

$$\widehat{\text{ATT}}_{\text{IPW}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)} \Delta Y_i$$

- $\hat{p}(X_i) = \widehat{\text{Pr}}(D_i = 1 \mid X_i)$ (propensity score)
- $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$
- If $\hat{p}(X_i) \rightarrow 1$ for a control unit, its weight $\rightarrow \infty$. **No overlap = no valid counterfactual.**

Abadie (2005): identification result. Estimate the pscore from a logit of D_i on X_i .

OR: outcome regression adjustment

$$\widehat{\text{ATT}}_{\text{OR}} = \frac{1}{n_T} \sum_{D_i=1} \left[\Delta Y_i - \hat{\mu}_0(X_i) \right]$$

$$\hat{\mu}_0(X_i) = \hat{E}[\Delta Y \mid D=0, X] \quad \text{estimated on controls only}$$

Key insight: the saturated TWFE (Spec C, which interacts $D_i \times X_i$) and OR are **algebraically identical** in the 2-period case. “Spec C is OR.” Both print \$1,770 on the LaLonde data.

DR: two chances to be right

$$\widehat{ATT}_{DR} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \underbrace{\frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)}}_{\text{IPW correction}} \cdot \underbrace{\left[\Delta Y_i - \hat{\mu}_0(X_i) \right]}_{\text{OR residual}}$$

$\hat{\mu}_0$ correct	$\hat{p}$ correct	Consistent?
✓	✓	✓
✓	×	✓
×	✓	✓
×	×	No

“Doubly robust” $\neq$ doubly unbeatable. Sant’Anna–Zhao (2020): `drdid` in Stata.

The LaLonde benchmark: we know the truth

- **LaLonde (1986)**: National Supported Work Demonstration — job training RCT. RCT estimate $\approx$ \$800.
- **LaLonde**: replaced randomized controls with CPS observational data. Re-ran every evaluator method. Results ranged from **—\$16,000 to +\$3,000**.
- **Dehejia & Wahba (2002)**: shorter sample requiring two pre-years. RCT benchmark on this sample: **\$1,794**.

NSW treated + CPS controls · Pre = 1975 · Post = 1978

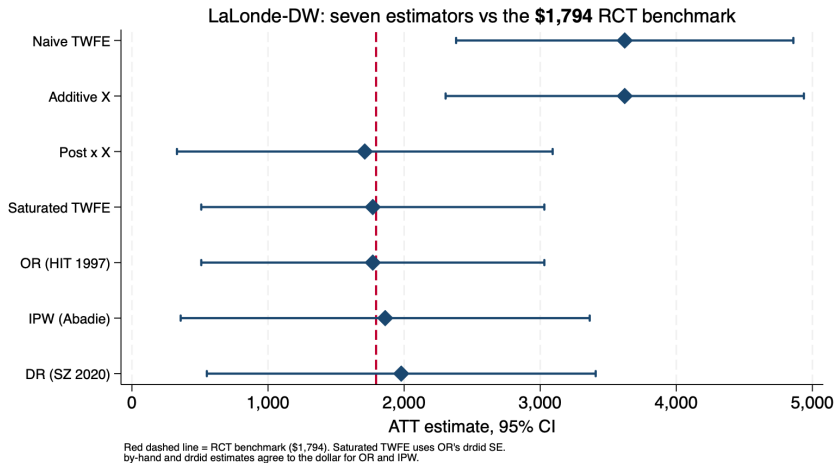
The benchmark exists. Which specs recover it?

How the specs do against the \$1,794 benchmark

Estimator	ATT	vs benchmark
Naive TWFE (no covariates)	\$3,621	+\$1,827
Additive X	\$3,621	+\$1,827
Post $\times X$ (Spec B)	\$1,711	-\$83
Saturated TWFE (Spec C)	\$1,770	-\$24
OR by hand / drdid reg	\$1,770	-\$24
IPW (drdid ipw)	\$1,861	+\$67
DR (drdid dripw)	\$1,979	+\$185

*Additive X : unit FE absorbs the level difference. The trend bias remains. **No help at all.***

All seven estimators vs the \$1,794 benchmark



Covariates are not robustness checks — they are design

- People say: “if adding covariates changes results, be suspicious.”
- That gets it backwards.
- We **know** the effect is \$1,794.
- Without covariates, or with additive covariates, we get \$3,621 — *wrong*.
- The correct specs recover \$1,770–\$1,979 — *right*.

Covariates fix broken parallel trends. Changing your result is not a bug. It is the design working.

The question is never

“did I control for X ?”

The question is always

“did I get the model right?”
